

Euclid's Elements — Book XI

Proposition 19

Formal Reconstruction — Technical Documentation

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CGJTEAMLAB

If two planes cutting one another be at right angles to any plane, then their common section will also be at right angles to the same plane.

1 Introduction

Proposition XI.19 is one of the most informative comparison points in the current Book XI development. The project now contains two production proofs of the same theorem:

```
CGJteamLab/Proposition11_19.lean
CGJteamLab/Wyler/Proposition11_19.lean
```

with main theorems

```
euclid_proposition_11_19
euclid_proposition_11_19_wyler
```

The first is the direct Hilbert 3D proof. The second is the Wyler reconstruction retained deliberately as a second proof route and as a laboratory for the passage from E^3 to E^4 .

The comparison is especially useful because XI.19 contains two logically different problems.

First, there is a global incidence problem: if two planes ρ, σ , both perpendicular to a third plane π , cut one another, then their traces on π must meet. The existence of this triple point is not a local right-angle fact; it is forced by the spatial parallel theory through XI.6 and XI.15.

Second, once a triple point

$$D \in \rho \cap \sigma \cap \pi$$

has been obtained, the remaining proof is local and neutral. XI.Def.4 supplies one normal to π inside each of the two perpendicular planes, XI.13 identifies the two normals, and exact plane intersection identifies that common normal with the common section of ρ and σ .

Both routes use the same neutral definition of perpendicular planes from

```
CGJteamLab/Hilbert3DPlanePerpendicular.lean
```

Thus, as in XI.18, the comparison concerns proof architecture rather than different theorem statements.

2 Euclid's Statement

Euclid states Proposition XI.19 as follows:

If two planes cutting one another be at right angles to any plane, then their common section will also be at right angles to the same plane.

Let ρ and σ be distinct planes which cut one another, and let π be a third plane. The hypotheses are

$$\rho \perp \pi, \quad \sigma \perp \pi.$$

If

$$t = \rho \cap \sigma,$$

then the conclusion is

$$t \perp \pi.$$

The formal conclusion contains more information than the bare statement that some normal to π lies in both planes. It returns an exact common section t and proves that this very line is perpendicular to π .

3 Classical Configuration

The configuration consists of three planes:

- the reference plane π ;
- a plane ρ perpendicular to π ;
- a plane σ perpendicular to π .

The two cutting planes have common section

$$t = \rho \cap \sigma.$$

Their traces on π are

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi.$$

The decisive point is

$$D = r \cap s.$$

Once this point is known,

$$D \in \rho \cap \sigma \cap \pi.$$

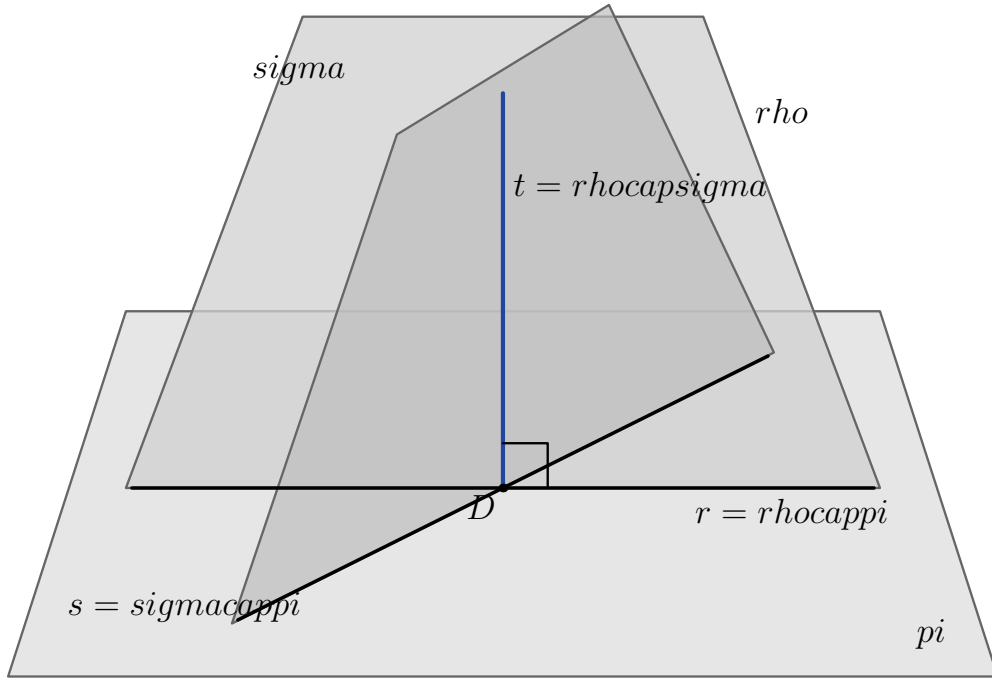


Figure 1: The classical configuration of Proposition XI.19. The planes ρ and σ are both perpendicular to the reference plane π . Their traces $r = \rho \cap \pi$ and $s = \sigma \cap \pi$ meet at D , and the common section $t = \rho \cap \sigma$ passes through the same point. The conclusion is $t \perp \pi$.

4 Euclid's Classical Proof

Euclid lets the two perpendicular planes cut one another in a straight line. At the point where the relevant traces on the reference plane meet, he constructs in each perpendicular plane a line perpendicular to the corresponding trace.

By XI.Def.4 each of those constructed lines is perpendicular to the reference plane. Since two straight lines perpendicular to the same plane at the same point must coincide, the two constructed normals are the same line.

That common normal lies in both cutting planes, hence it is their common section. Therefore the common section is perpendicular to the reference plane.

The printed proof makes the triple point visually natural. In formal geometry that point is not free. The statement

$$r \cap s \neq \emptyset$$

is the main global incidence theorem inside the proof.

5 Classical Dependency Pattern

At the level of the classical diagram the argument has the compact shape

$$\begin{array}{c}
 \rho \perp \pi, \quad \sigma \perp \pi, \quad \rho \text{ cuts } \sigma \\
 \Downarrow \\
 D \in \rho \cap \sigma \cap \pi \\
 \Downarrow \\
 g \subset \rho, \quad h \subset \sigma, \quad g \perp \pi, \quad h \perp \pi \text{ at } D \\
 \Downarrow \\
 \text{uniqueness of the normal at } D \\
 \Downarrow \\
 g = h = \rho \cap \sigma \\
 \Downarrow \\
 (\rho \cap \sigma) \perp \pi.
 \end{array}$$

The formal development separates the first arrow from the remainder. That separation is mathematically significant: existence of D is the only global Euclidean part of XI.19.

6 Shared XI.Def.4 Structure

Both formal proofs use the neutral predicates

```
HilbertSpacePlanesPerpendicularAlong Geo rho pi r
HilbertSpacePlanesPerpendicular Geo rho pi
```

from

```
CGJteamLab/Hilbert3DPlanePerpendicular.lean
```

A certificate that ρ is perpendicular to π contains an exact common section r and the XI.Def.4 elimination rule:

$$g \subset \rho, \quad g \perp r \text{ at } D \implies g \perp \pi \text{ at } D.$$

XI.19 uses this rule twice, once in each of the two perpendicular planes.

This is why XI.18 and XI.19 must not use the modern existential definition

$$\rho \perp \pi \quad :\Longleftrightarrow \quad \exists g \subset \rho, g \perp \pi.$$

Such a definition would suppress the common-section geometry that is the actual content of both propositions.

7 What Is Genuinely Three-Dimensional Here?

XI.19 is spatial in several independent ways.

First, the theorem concerns perpendicularity of planes rather than merely perpendicularity of lines.

Second, three exact plane meets occur:

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi, \quad t = \rho \cap \sigma.$$

In E^3 , each of these meets is a line.

Third, the existence of the point

$$D \in \rho \cap \sigma \cap \pi$$

requires a global spatial argument. If the traces $r, s \subset \pi$ were disjoint, they would be parallel. Normals erected inside ρ and σ would then supply a second pair of parallel lines. XI.15 would force

$$\rho \parallel \sigma,$$

contradicting the given common point of the cutting planes.

Finally, after the triple point is obtained, XI.13 is a spatial uniqueness theorem: two lines perpendicular to the same plane at the same point coincide.

8 Formal Statements

8.1 Direct Hilbert 3D theorem

The direct Hilbert production theorem is

```

theorem euclid_proposition_11_19
  [H : HilbertIncidence Geo]
  [HilbertPlaneIncidence Geo]
  [S : HilbertSpacePrimitive Geo]
  [HSI : HilbertSpaceIncidence Geo]
  [_HSO : HilbertSpaceOrder
    (Geo := Geo) (H := H) (S := S)]
  [HSC : HilbertSpaceCongruence

```

```

(Geo := Geo) (H := H) (S := S)]
[HSE : HilbertSpaceEuclidean Geo]
(rho sigma pi : S.Plane)
(K : Geo.Point)
(hRhoSigma : Ne rho sigma)
(hKrho : S.OnPlane K rho)
(hKsigma : S.OnPlane K sigma)
(hRhoPerp :
  HilbertSpacePlanesPerpendicular Geo rho pi)
(hSigmaPerp :
  HilbertSpacePlanesPerpendicular Geo sigma pi) :
exists D : Geo.Point,
exists t : Geo.Line,
  S.OnPlane D pi /\
  H.OnLine D t /\
  HilbertLineInPlane Geo t rho /\
  HilbertLineInPlane Geo t sigma /\
  (forall X : Geo.Point,
    (S.OnPlane X rho /\ S.OnPlane X sigma) <=>
    H.OnLine X t) /\
  HilbertLinePerpendicularPlaneAt Geo t pi D

```

Its production file is

```
CGJteamLab/Proposition11_19.lean
```

8.2 Wyler theorem

The Wyler theorem has the same public type:

```

theorem euclid_proposition_11_19_wyler
[H : HilbertIncidence Geo]
[HilbertPlaneIncidence Geo]
[S : HilbertSpacePrimitive Geo]
[HSI : HilbertSpaceIncidence Geo]
[_HSO : HilbertSpaceOrder
  (Geo := Geo) (H := H) (S := S)]
[HSC : HilbertSpaceCongruence
  (Geo := Geo) (H := H) (S := S)]
[HSE : HilbertSpaceEuclidean Geo]
(rho sigma pi : S.Plane)
(K : Geo.Point)
(hRhoSigma : Ne rho sigma)
(hKrho : S.OnPlane K rho)
(hKsigma : S.OnPlane K sigma)
(hRhoPerp :
  HilbertSpacePlanesPerpendicular Geo rho pi)
(hSigmaPerp :
  HilbertSpacePlanesPerpendicular Geo sigma pi) :
exists D : Geo.Point,
exists t : Geo.Line,
  S.OnPlane D pi /\
  H.OnLine D t /\

```

```

HilbertLineInPlane Geo t rho /\
HilbertLineInPlane Geo t sigma /\
(forall X : Geo.Point,
  (S.OnPlane X rho /\ S.OnPlane X sigma) <=>
    H.OnLine X t) /\
HilbertLinePerpendicularPlaneAt Geo t pi D

```

Its production file is

```
CGJteamLab/Wyler/Proposition11_19.lean
```

The hypotheses

```

hRhoSigma : Ne rho sigma
hKrho      : S.OnPlane K rho
hKsigma    : S.OnPlane K sigma

```

formalize Euclid's phrase that the two planes cut one another.

9 Direct Hilbert 3D Reconstruction

The direct production file imports

```

CGJteamLab.Hilbert3DPlanePerpendicular
CGJteamLab.Proposition11_6
CGJteamLab.Proposition11_13
CGJteamLab.Proposition11_15

```

and contains the complete direct proof, including its own local normal construction and its own common-point core.

9.1 Stage H1: extract the two traces

From the two perpendicular-plane certificates the proof obtains exact section lines

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi.$$

Their containments are explicit:

$$r \subset \rho, \quad r \subset \pi, \quad s \subset \sigma, \quad s \subset \pi.$$

9.2 Stage H2: force the traces to meet

Assume for contradiction that r and s do not meet.

Because both lie in π ,

$$r \parallel s.$$

Choose points

$$R \in r, \quad Q \in s.$$

Since the traces are disjoint,

$$R \neq Q.$$

The direct helper

```
hilbert_XI19_planesPerpendicularAlong_normal_exists
```

uses XI.Def.4 to produce

$$g \subset \rho, \quad g \perp \pi \text{ at } R,$$

and

$$h \subset \sigma, \quad h \perp \pi \text{ at } Q.$$

Then direct XI.6 gives

$$g \parallel h.$$

The pairs (r, g) and (s, h) are intersecting line pairs inside ρ and σ , respectively. The proof verifies explicitly that the four lines cannot all lie in one plane; otherwise uniqueness of the plane through two intersecting lines would imply

$$\rho = \sigma.$$

Direct XI.15 now yields

$$\rho \parallel \sigma.$$

But the hypothesis supplies

$$K \in \rho \cap \sigma.$$

Contradiction. Therefore r and s meet.

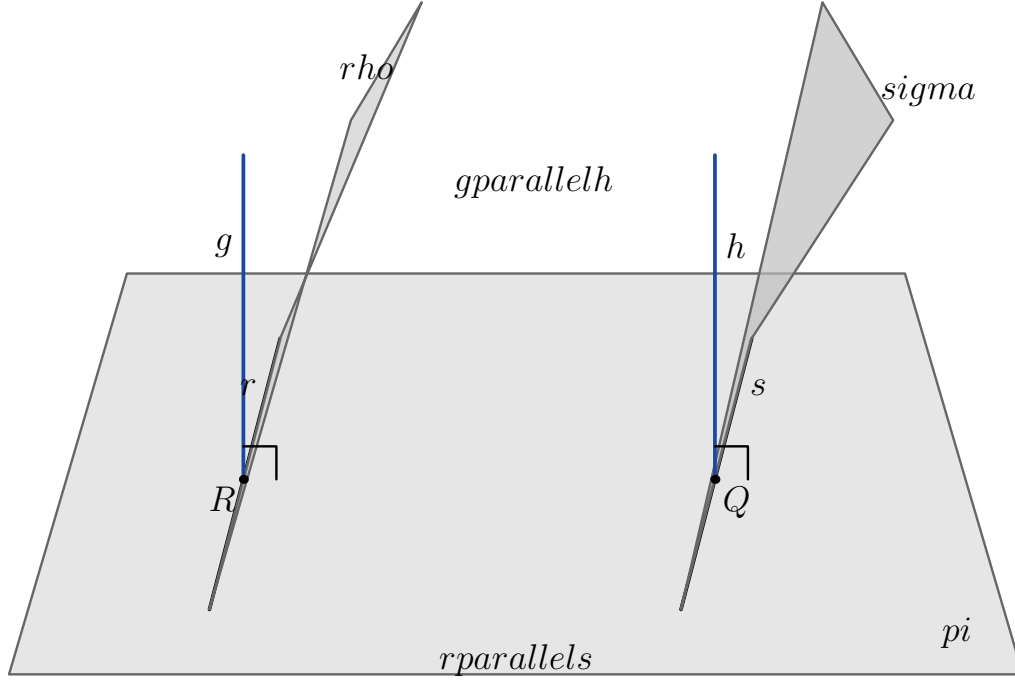


Figure 2: The incidence step hidden by the classical picture. If the traces $r = \rho \cap \pi$ and $s = \sigma \cap \pi$ were disjoint, then $r \parallel s$. Normals $g \subset \rho$ and $h \subset \sigma$ to π are parallel by XI.6. XI.15 would then force $\rho \parallel \sigma$, contradicting the common point K . Hence r and s meet at the required triple point D .

9.3 Stage H3: direct common-point core

Let D be the intersection point of r and s . Then

$$D \in \rho, \quad D \in \sigma, \quad D \in \pi.$$

The direct theorem

`euclid_proposition_11_19_common_point`

now handles the local part.

First, XI.Def.4 produces

$$g \subset \rho, \quad g \perp \pi \text{ at } D,$$

and

$$h \subset \sigma, \quad h \perp \pi \text{ at } D.$$

Direct XI.13 gives

$$g = h.$$

Next, exact plane intersection gives

$$t = \rho \cap \sigma.$$

Since the common normal lies in both ρ and σ , a second point on that normal lies on the exact meet t . Line uniqueness then gives

$$g = t.$$

Therefore

$$t \perp \pi \quad \text{at } D.$$

9.4 Stage H4: local planar construction

The lowest direct helper is

```
hilbert_XI19_perpendicular_line_through_point_in_plane
```

It enters

```
PlaneGeo Geo rho
```

and erects a perpendicular to the section line at a specified section point. The result is transported back to the ambient geometry and consumed by the XI.Def.4 elimination rule.

Thus even the direct Hilbert proof has a clear local 2-dimensional core inside an ambient 3-dimensional argument.

10 Wyler Reconstruction

The Wyler production theorem imports

```
CGJteamLab.Wyler.Proposition11_19_common_point_wyler
CGJteamLab.Wyler.Proposition11_6
CGJteamLab.Wyler.Proposition11_15
```

The visible proof architecture is intentionally parallel to the Hilbert route.

10.1 Stage W1: global triple-point layer

The Wyler proof extracts the exact traces

$$r = \rho \cap \pi, \quad s = \sigma \cap \pi$$

from the same neutral XI.Def.4 certificates.

Assuming the traces disjoint gives

$$r \parallel s.$$

At points $R \in r$, $Q \in s$, the helper

```
hilbertSpacePlanesPerpendicularAlong_normal_exists_wyler
```

supplies normals

$$g \subset \rho, \quad h \subset \sigma$$

to π .

Wyler XI.6 gives

$$g \parallel h,$$

and Wyler XI.15 forces

$$\rho \parallel \sigma,$$

contradicting the common point K .

Hence the triple point D exists.

10.2 Stage W2: isolated common-point core

The local result is isolated in

```
CGJteamLab/Wyler/Proposition11_19_common_point_wyler.lean
```

with theorem

```
euclid_proposition_11_19_common_point_wyler
```

At the given triple point, XI.Def.4 supplies one normal in each cutting plane; XI.13 identifies the normals; exact flat intersection identifies that normal with the common section of ρ and σ .

The lower normal-extraction machinery belongs to

```
CGJteamLab/Wyler/HilbertWylerPlaneNormal.lean
```

The Wyler route therefore makes the separation between

global existence of D

and

local identification of the common normal

an explicit module boundary.

11 Hilbert versus Wyler: Proof Diff

XI.19 is particularly useful because the two proofs are mathematically close but architecturally distinct.

11.1 Shared skeleton

Both proofs have the same high-level graph:

$$\begin{array}{c} r = \rho \cap \pi, \quad s = \sigma \cap \pi \\ \Downarrow \\ \text{assume } r \cap s = \emptyset \\ \Downarrow \\ r \parallel s \\ \Downarrow \\ \text{construct normals } g, h \text{ to } \pi \\ \Downarrow \\ \text{XI.6 : } g \parallel h \\ \Downarrow \\ \text{XI.15 : } \rho \parallel \sigma \\ \Downarrow \\ \perp \\ \Downarrow \\ D \in \rho \cap \sigma \cap \pi \\ \Downarrow \\ \text{XI.Def.4 + XI.13} \\ \Downarrow \\ (\rho \cap \sigma) \perp \pi. \end{array}$$

11.2 Direct Hilbert realization

The direct route keeps the complete argument in the main Book XI hierarchy:

```
Proposition11_19
-> Proposition11_6
-> Proposition11_13
-> Proposition11_15
```

and defines direct local helpers inside `Proposition11_19.lean`.

This is the natural proof in Hilbert’s E^3 language.

11.3 Wyler realization

The Wyler route factors the proof more aggressively:

```
Wyler.Proposition11_19
  -> Wyler.Proposition11_19_common_point_wyler
  -> Wyler.HilbertWylerPlaneNormal
  -> Wyler.Proposition11_6
  -> Wyler.Proposition11_15
```

The global incidence stage remains visible in the final theorem, while the local common-point argument is promoted to a separate neutral layer.

11.4 The mathematical difference

The difference is not the final theorem and not even the high-level proof. The difference is where the geometry is cut into reusable interfaces.

Hilbert says, in effect:

solve XI.19 directly in the native E^3 language.

Wyler says:

separate exact flat configuration from metric identification.

This is precisely the comparison needed before moving to E^4 .

12 Ambient Space and PlaneGeo

The public theorem is entirely ambient:

```
rho sigma pi : S.Plane
r s t        : Geo.Line
D K          : Geo.Point
```

The induced plane geometry appears only in the local normal construction. Given a section line $r \subset \rho$ and a point $D \in r$, the proof enters

```
PlaneGeo Geo rho
```

to erect a perpendicular to r at D .

The following steps remain genuinely ambient:

- exact intersection of two planes;
- spatial parallelism of lines;
- XI.6 for normals to a plane;
- XI.15 for parallel planes;
- XI.13 for uniqueness of a normal to a plane;
- identification of a normal line with an exact common section.

13 Hidden Formal Data

13.1 The cutting planes are distinct

The hypothesis

```
hRhoSigma : Ne rho sigma
```

is required because an exact common section line of two identical planes would not be determined.

13.2 The cutting planes have a common point

The data

```
K : Geo.Point
hKrho   : S.OnPlane K rho
hKsigma : S.OnPlane K sigma
```

formalize Euclid's phrase that ρ and σ cut one another.

13.3 The triple point is derived

The point D is not an input. The proof derives

$$D \in \rho \cap \sigma \cap \pi$$

from the trace argument using XI.6 and XI.15.

This is the most important incidence fact hidden by the classical picture.

13.4 The common section is exact

The result includes

```
forall X : Geo.Point,
  (S.OnPlane X rho /\ S.OnPlane X sigma) <->
    H.OnLine X t
```

Thus t is not merely a line contained in both planes. It is the full point-set intersection.

13.5 Normal extraction is constructive

At a specified point of the XI.Def.4 section, the proof first constructs a line inside the carrier plane perpendicular to that section, then applies the XI.Def.4 elimination rule to upgrade the line to a normal of the reference plane.

This local construction is explicit in both proof routes.

14 Relationship with Earlier Book XI Propositions

XI.6. Two lines perpendicular to the same plane at distinct points are parallel. This creates the second parallel pair needed in the contradiction argument.

XI.13. The perpendicular to a plane at a prescribed point is unique. This is the central local theorem after the triple point has been obtained.

XI.15. Two carrier planes containing corresponding pairs of parallel lines are parallel under the noncoplanarity hypothesis. This is the decisive global Euclidean step.

XI.18 and XI.Def.4. XI.18 established the source-faithful perpendicular-plane interface now shared by both routes. XI.19 uses that interface as an elimination rule rather than introducing a new definition.

XI.19 therefore closes a short chain:

$$\text{XI.18} \longrightarrow \text{usable XI.Def.4 plane perpendicularity} \longrightarrow \text{XI.19.}$$

15 Relationship with Hilbert's Axiom Groups

The local common-point core uses the neutral synthetic geometry:

- Group I incidence;
- Group II order in the planar perpendicular construction;
- Group III congruence and right-angle structure.

The full theorem additionally requires

because the global triple-point argument invokes XI.15.

Thus Group IV is localized to the existence layer:

$$\text{if the traces do not meet} \implies \rho \parallel \sigma.$$

Once D exists, the identification of the common section with the normal to π is neutral.

16 Classical DAG and the Two Formal DAGs

16.1 Euclid

$$\begin{array}{c} \rho \perp \pi, \quad \sigma \perp \pi, \quad \rho \text{ cuts } \sigma \\ \Downarrow \\ D \in \rho \cap \sigma \cap \pi \quad \text{visible in the configuration} \\ \Downarrow \\ \text{two normals to } \pi \text{ at } D \\ \Downarrow \\ \text{uniqueness} \\ \Downarrow \\ (\rho \cap \sigma) \perp \pi. \end{array}$$

16.2 Direct Hilbert 3D

$$\begin{array}{c} \rho \perp \pi, \quad \sigma \perp \pi, \quad K \in \rho \cap \sigma, \quad \rho \neq \sigma \\ \Downarrow \\ r = \rho \cap \pi, \quad s = \sigma \cap \pi \\ \Downarrow \\ r \cap s = \emptyset \Rightarrow r \parallel s \\ \Downarrow \\ \text{hilbert_XI19_planesPerpendicularAlong_normal_exists} \\ \Downarrow \\ \text{XI.6 : } g \parallel h \\ \Downarrow \\ \text{XI.15 : } \rho \parallel \sigma \\ \Downarrow \\ \perp \\ \Downarrow \\ \exists D \in \rho \cap \sigma \cap \pi \\ \Downarrow \\ \text{euclid_proposition_11_19_common_point} \\ \Downarrow \\ \text{XI.13 + exact meet} \\ \Downarrow \\ (\rho \cap \sigma) \perp \pi. \end{array}$$

16.3 Wyler 3D

$$\begin{array}{c}
\rho \perp \pi, \quad \sigma \perp \pi, \quad K \in \rho \cap \sigma, \quad \rho \neq \sigma \\
\Downarrow \\
r = \rho \cap \pi, \quad s = \sigma \cap \pi \\
\Downarrow \\
r \cap s = \emptyset \Rightarrow r \parallel s \\
\Downarrow \\
\text{hilbertSpacePlanesPerpendicularAlong_} \\
\text{normal_exists_wyler} \\
\Downarrow \\
\text{Wyler XI.6 : } g \parallel h \\
\Downarrow \\
\text{Wyler XI.15 : } \rho \parallel \sigma \\
\Downarrow \\
\perp \\
\Downarrow \\
\exists D \in \rho \cap \sigma \cap \pi \\
\Downarrow \\
\text{euclid_proposition_11_19_common_point_wyler} \\
\Downarrow \\
\text{XI.13 + exact flat meet} \\
\Downarrow \\
(\rho \cap \sigma) \perp \pi.
\end{array}$$

The two DAGs are close at proposition level. Their difference lies in the modularization of the lower geometry.

17 E^4 Transfer Audit

XI.19 reveals a stronger dimension dependence than XI.18.

17.1 1. Purely local step

Once a point D lies in all relevant carriers, the construction of a normal direction can be reduced to a suitable 2-dimensional normal section.

The local pattern

$$\text{construct normal directions at } D \quad + \quad \text{use uniqueness}$$

is a good candidate for dimension-independent packaging.

17.2 2. Genuinely E^3 incidence step

In E^3 ,

$$\rho \cap \pi, \quad \sigma \cap \pi, \quad \rho \cap \sigma$$

are lines when the corresponding planes are distinct and intersect.

This fact is built deeply into the statement of XI.19. In E^4 , two 2-planes need not meet in a line, and two hyperplanes meet generically in a 2-flat. Therefore the carrier dimensions in the theorem cannot simply be copied.

17.3 3. Wyler replacement

The Wyler organization suggests replacing fixed words such as “intersection line” by typed flat meets with explicit dimension data.

The transferable pattern is

flat meet $\longrightarrow$ common lower-dimensional carrier $\longrightarrow$ metric normal identification.

This is more promising for E^4 than a direct transcription of the three-dimensional statement.

17.4 4. Expected E^4 analogue

A useful higher-dimensional analogue should likely be formulated in terms of codimensions rather than the names “line” and “plane”.

The XI.19 pattern suggests investigating statements of the form:

if two suitable flats are both orthogonal to a reference flat and meet with the required dimension, then their common meet inherits an orthogonality property determined in an appropriate normal 2-flat.

The exact formulation depends on the chosen E^4 flat and orthogonality interfaces and is not asserted here as an already formalized theorem.

17.5 5. Missing E^4 lemmas

XI.19 points to at least three higher-dimensional interfaces:

dimension formula for flat meets],

normal-space / normal-2-flat construction],

and

higher-dimensional analogue of XI.15].

The last item is particularly important. In the E^3 proof, XI.15 turns two pairs of parallel lines into parallelism of the carrier planes. In E^4 , the correct statement must know the dimensions and spanning relations of the corresponding flats.

Thus XI.19 is a stronger diagnostic for the E^4 project than XI.18: the main difficulty is no longer merely replacing one intersection line by a higher-dimensional meet; the global parallel-carrier argument itself must be reformulated dimensionally.

18 Structural Lesson

XI.19 shows why the project should retain the direct Hilbert and Wyler routes side by side.

The direct Hilbert proof demonstrates that the theorem is natural and manageable in Hilbert's synthetic E^3 language. Nothing about the three-dimensional argument requires abandoning that language.

The Wyler proof, however, makes a different feature visible: the theorem can be decomposed into an exact incidence layer and a local metric layer. That decomposition is precisely what becomes valuable when the dimensions of intersections change in E^4 .

The research object is therefore the proof difference:

$$\text{Hilbert } E^3 \leftrightarrow \text{Wyler } E^3 \implies \text{candidate dimension-free interfaces.}$$

For XI.19 the most important candidate is the replacement of the rigid three-dimensional pattern

$$\text{two line traces} + \text{two parallel normals} \Rightarrow \text{parallel planes}$$

by a dimension-aware statement about flats and their spanning directions.

19 Audit Status

The current project contains the production theorems

```
Geometry.euclid_proposition_11_19
Geometry.euclid_proposition_11_19_wyler
```

together with the neutral XI.Def.4 module

```
CGJteamLab/Hilbert3DPlanePerpendicular.lean
```

and both Book XI routes are included through XI.19 in the main project imports.

The direct XI.19 theorem and the complete project were reported to compile cleanly after the XI.18/XI.19 integration.

No new `#print axioms` result is claimed in this documentation revision.

20 Conclusion

Proposition XI.19 has the compact classical form

$$\rho \perp \pi, \quad \sigma \perp \pi, \quad \rho \text{ cuts } \sigma \quad \Longrightarrow \quad (\rho \cap \sigma) \perp \pi.$$

The current formalization now contains two independent reconstructions.

Both first solve the global incidence problem of producing the triple point D . Both then solve the neutral local problem by extracting two normals to π , identifying them by XI.13, and identifying the resulting line with the exact common section of ρ and σ .

The direct Hilbert route keeps this argument in the native synthetic language of E^3 . The Wyler route factors the same mathematics into explicit flat and metric layers.

For the E^4 program, XI.19 therefore supplies a particularly useful lesson: the local normal argument appears transferable, while the dimensions of the meets and the XI.15-style global parallelism step are genuinely three-dimensional and must be reformulated rather than copied.